

Choosing an Effective Visual Transcript

Hello everyone, welcome to BITS Pilani digital. Today we will be continuing our learning journey in the course of data visualisation storytelling. We will start a new module today, taxonomy of data visualisation methods.

Today's lesson we are going to see two concepts, one is how to choose an effective visual and the design concepts to help you design better visualisations. There are about 150 plus different types of visualisations that you can readily use in any software, let us say Excel and if you take in fact Power BI and tablet is even more higher, but however most of the times you might need about a dozen of visuals to convey the common information. In practise these dozen visualisations would serve 90% of your cases, the rest 5% being the custom visuals where you have to design or use specific visualisations to drive home a story, but in the majority of cases knowing a couple of or a handful of visualisations thoroughly will help you design and deliver effective visualisations.

So therefore data visualisations though might require high-end software can be done using simple tools, but only if you should know how to effectively use them. Now let us see what are these basic visualisations and how can you effectively use them, because it is important to use the right visualisation for the right purpose and the choice of the visual therefore becomes very important. And what are these visuals that are generally used? There are simple text, table or heat map, then you have scatter plot, line and slope graph, bar vertical, stacked, horizontal, then you have waterfall and square area.

These are different types of graphs and you have to understand why a particular graph or a visualisation is best suited for a given purpose. To start with simple text, you might ask what is simple text to do with visualisation? In fact most cases doing a simple text based presentation of data or something that you want to highlight is the best way to communicate the information. Where there is no need to show a visual, please don't show any visual, you can simply convey the meaning with text and it can be easily observed.

And by text I mean it should not be a long phrase but a small data point or a phrase that you can highlight and get across the information. Let me tell you through this an example. Consider the bar graph that is in the centre.

It shows children with traditional stay-at-home mother and the percentage has come down 41 to 20 percent within 1970 to 2012 indicating that the stay-at-home mothers are falling because many of them are moving to workforce. You have two bars and somebody has to apply cognitive information or has to process saying that it has fallen from 41 to 20. Instead you can simply say currently the stay-at-home mother ratio is 20% as compared to 41% in 1970.

The text on the right hand side would capture this succinctly and deliver the impact. You might be using the same space as the visual but you are just replacing that with

the text it can give the same impact. And also you might recall many a times in the within the visualisations we use text inserts to create this impact of delivering a particular information.

Text becomes an undeniable part of all visualisation exercises though it might not look per se as a visualisation. And you can always use gestalt principles of visual perception to improve the content delivery of the text. You can use bold, you can use colour, you can use different shape to give this impacts to text.

Next we will see tables and heatmaps. Tables interact with our verbal system. When we present data in table we are actually conveying a verbal information to the audience.

The audience intuitively try to read the tables either horizontally or vertically unless they are guided particularly to go either column to column or row to row. They would intuitively read the table and form conclusions they interact with our verbal system very strongly. And this can be used to put across very simple information without trying to give a particular insight into the data.

However if you want to drive insights within tables you can use heatmaps and use the saturated or graded colours to drive home the concentrations or compositions or direction of the data that can be done using heatmaps. In today's Excel software you have conditional formatting which can easily help somebody to create a heatmap. And of course in Python or Power BI Tableau you have softwares that create heatmaps once you give the data.

So the heatmaps are an interesting way to present the distributional composition of the data within the table. So you still retain the verbal element but are trying to convey a particular point to your audience. Tables are very common in data visualisation exercises and report writing like simple text.

Tables are also very potent source for delivering information when it is very simple. Next we go to graphs which forms the main summons of sensor the data visualisation exercises. There are four types of graphs that generally are used.

They are point graphs, bar graphs, line graphs and area graphs. Each of them serve different purposes. We have to be very clear which type of graph we are using for the purpose at hand.

Let us take all these four types individually. First graph is the scatterplot. It is the graph type of point.

The elements in this graphs are points. So when we try to understand relationship between two variables scatterplot is something that is most ideal. Let us say in the earlier lessons we have tried to understand the relationship between average per booth voters versus the voter turnout participation.

In this case we did not know the priori we did not know the relationship. By using a scatterplot we could understand there is a slight negative relationship. For exploring relationships scatterplot is very ideal.

It uses X and Y data and represents the data as a point. The extensions of a scatterplot are the bubble chart where we can represent the extension of a scatterplot is a bubble chart where the size of the bubble indicates the quantity of one of the variables or the third variable that you can impose and in fact you can use colour shape of different sizes to add more detail to the scatterplot. Scatterplots become very interesting when you are doing exploratory analysis when you do not know the inherent relationships you can use scatterplots to guide you through this process.

Now we will talk about another type of graphs called lines. Here the object in the graph is a line indicating continuity. Following the gestalt principle whenever there is a continuous shape it is intuitive for the human mind to think these are continuous variables.

Therefore whenever you are plotting lines it is inherently for continuous variables. However you can use a slope graph for plotting the changes in categorical variables as shown in the right side of the slide. Here an employable feedback sample is considered where you have different categories of the feedback and the changes between those two types feedback between 2014 and 15.

Though these are categorical variables the change can be shown as a slope graph as simply connecting the two points between the two time periods. This is very useful when you want to understand the rate of change between time periods and this can be extended. As opposed to this you have lines which are done on continuous variables you might have a single series line representing one variable or multi-series line again.

Caution has to be avoided that if you clutter the graph you might lose to convey the information. You can effectively use the other gestalt principles or colour coding to effectively deliver information using line graphs. Then we move to the most important bar type which is commonly used to convey information to public.

This is bar graph where the bar the graph elements are individual bars. Now you have primary horizontal and vertical bar depending upon whether the bar is stacked horizontally or vertically. So these are used to show comparison between two small categories and these are very intuitive because the length of the bar indicates the quantity of the variable and you can make comparisons of the variable over a period of time or across categories.

This can be built into the graphs. There is one more type of bar graph called the stacked bar graph which is very useful to see composition of the variable across categories and within the categories. Stacked bar graph is a very powerful present data and composition in the same visual.

For example let us take the graph at the bottom and stacked 100% bar. It can be used as an alternative to the pie diagram. For example here in this case we are looking at a survey result.

The survey has been done from four different or five different units and there are four five components of the survey. You can see that by dividing the composition into four five categories the graph is clearly able to show that from survey A there are people who have strongly agreed or very high as compared to survey E where people who have disagreed are very high. So these kind of insets can be readily drawn from a stacked graph.

For all the bar graphs you have should have quantitative data to present them so that you can make comparisons readily. Now we will move to a special case of bar graphs called waterfall graph. This is an interesting case of bar graph where it shows the sum of movement from point A to point B as some components.

Let me explain this with an example. You have an initial value and at the end of the time period it moves to some other value. In this case we are talking about total number of employees in a team.

Initially there were 100 people and at the end of the time period it has moved to 116. If you just see those two points you would come to a conclusion the number of employees have increased by 16 pp but it doesn't talk about how this change has come. There are three ways in this changes has come or four ways.

There have been new hires, there have been transfers, there have been transfers to the department, transfers out of the department and there are exits. When we consider all these things together the waterfall diagram is the best to depict this. I am starting with 100 people, I have recruited 30 people, 8 people have been transferred to the department, 12 people have left the department because of transfers, 10 people have exited the department, final count is 116.

This will help us to understand how a variable is moving from point A value point A to point B and what are the components that are causing these changes. This is very essential and useful in presenting the profitability results of a company considering how profit revenue is becoming profit or how revenue is increasing between two time periods. You can take account expenditures and show the positive growth from revenue and negative growth from expenditures and clearly present whether the company is doing good or bad over a period of time.

Waterfall chart is inbuilt in some of the software's otherwise there are plug-ins that are available that you can use to deliver the waterfall chart. Finally we will see one more graphs called area. Here the idea is to present the composition in an area let us say a rectangle or a square and the higher is the share of the area occupied greater is the value of that particular component or variable.



In our case we are talking about population of Indian states. As you know Uttar Pradesh is the highest population state. As you know Uttar Pradesh has the highest population in India. It occupies the major share of area in the particular rectangle is one more rectangle. The other rectangle is followed by Bihar, Maharashtra, West Bengal and the smaller population states have lesser rectangular space in the graph. This shows that an area chart can be shown as an alternative composition and it is very intuitive because it occupies larger space you can simply see this has larger value right.

So area chart is one more way to present the composition diagrams and we also have similar type of area chart called square area where the entire squares represent the number of squares occupied by a variable indicate the composition within the data and this is an interesting graphic especially when you want to present composition across time period. You can simply present as a squares two squares and show how the composition has changed over a period of time. This can readily give ah interpretations to the audience.

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